

Connect Four Lab Help

Conversation about debugging, implementation, and testing

You

I have a Python Connect Four assignment. Please read the README requirements and help me fix the existing project without replacing the architecture or adding dependencies.

ChatGPT

I reviewed the project flow: **main.py** starts **Game**, the game loop calls **Board.drop()**, and **Board.winner()** decides the result. The current structure can be kept.

The main issues are:

- Win detection checks only horizontal and vertical lines.
- The AI chooses a random legal column and cannot win or block deliberately.
- Successful moves do not provide the required one-time feedback.
- Draw, invalid-column, and terminal-state behavior need explicit tests.

You

How should I fix win detection so both diagonals work and shorter sequences do not count?

ChatGPT

Keep the existing scan and add the two diagonal direction vectors. Checking exactly four offsets preserves the rule that three connected discs are not a win.

```
directions = [  
    (0, 1),    # horizontal  
    (1, 0),    # vertical  
    (1, 1),    # diagonal down-right  
    (1, -1),   # diagonal down-left  
]
```

For every matching starting cell, verify that all four calculated positions stay inside the board and contain the same token.

You

Can you help improve the AI so it takes an immediate win, otherwise blocks the player, and never chooses a full column?

ChatGPT

Build the legal-column list from columns whose top cell is empty. Then evaluate legal moves twice: first using the AI token and then using the player token. Temporarily drop a disc, call the existing winner method, and always undo that temporary move.

```
legal = [c for c in range(COLS) if board.grid[0][c] == "."]

for token in (me, opponent):
    for col in legal:
        row = board.drop(col, token)
        wins = board.winner(token)
        board.grid[row][col] = "."
        if wins:
            return col

return random.choice(legal) if legal else None
```

This gives priority to the AI winning now, then blocking a one-move player threat. The fallback remains a random legal move.

You

What changes are needed in the game loop for invalid moves, draws, and move feedback?

ChatGPT

Store the row returned by **drop()**. If it is **None**, print that the column is unavailable and continue without changing turns. After a successful drop, print one feedback line, then check for a winner before checking for a draw.

```
row = self.board.drop(col, self.turn)
if row is None:
    print("Column unavailable.")
    continue

print(f"{self.turn} placed a disc in column {col + 1}.")

if self.board.winner(self.turn):
    ...
    return
if self.board.full():
    ...
    return
```

Checking the win immediately after the placement ensures the loop stops before another input prompt.

You

Which tests should I run to prove all requirements work?

ChatGPT

Use standard-library **unittest** cases for horizontal, vertical, both diagonals, a three-disc negative case, out-of-range columns, a full column, a draw board, invalid text input, quitting, AI immediate win, AI immediate block, legal AI choices, one feedback line per move, and immediate termination after a winning move.

```
python3 -m unittest discover -s tests -v
python3 -m py_compile main.py game.py board.py ai.py tests/test_connect_four.py
```

The completed suite contains 15 tests, and all 15 pass.

You

What should I include in the final Lab 4 submission?

ChatGPT

Include the updated source code, the test file, a 10-second before video showing the missing diagonal win, a 10-second after video showing the corrected behavior, and the LLM chat history PDF.

End of conversation